

Linear systems and elimination

A sharp subset bound for worst-case normal GMRES

Difficulty: challenging **Importance:** interesting to the community**Status:** [Lean verified](#)

Historical rating rationale: Challenging reflects extending a discrete real approximation principle to complex spectra with a sharp constant; community impact concerns usable worst-case GMRES bounds for normal matrices.

Resolution: negative, 12 September 2026

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Theorem 1.1, proved in Sections 2-3 of the [manuscript](#) disproves the displayed universal inequality. With $\omega = e^{2\pi i/3}$, take the nine distinct nonzero points $L = \{\omega^a + 10^{-3}\omega^b : 0 \leq a, b \leq 2\}$ and $k = 4$. Then

$$M_4(L) = \frac{3003003000}{1001003001001} > \frac{299}{100000}, \quad \max_{S \subseteq L, |S|=5} M_4(S) < \frac{23}{10000}.$$

The ratio is therefore greater than $13/10 > 4/\pi$. Here $n = 9$ and $1 \leq 4 \leq n-2$, so all original assumptions and quantifiers are addressed by this counterexample. Theorem 1.2 (Sections 4-5) additionally proves that no finite dimension-independent replacement constant exists. The finite counterexample alone settles IE-16 negatively; no case of the original universal truth question remains open.

The complete proof passed a separate [independent Codex AI-agent informal audit](#). The supplied exact-arithmetic verifier passed all 126 subset checks, and reviewer-written rational checks passed independently. That dated informal audit was not external human peer review or formal verification. The separate Lean verification is recorded below. [Proof source](#) · [Exact verifier and certificates](#).

Lean proof and verification evidence — 2026-09-13

Formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA. Original mathematical proof and exact certificate: **Sidney Holden**, Center for Computational Biology, Flatiron Institute, Simons Foundation.

The [immutable complete Lean proof](#) uses Lean 4.33.1, [LeanCert 621a43d7cf21](#) and [Mathlib 0df444a360ea](#), with full pins in the [dependency manifest](#). The [project guide](#), [exact targets](#), [definitions](#) and [fifteen-contract boundary](#) record the precise source correspondence.

`NLA.IE16.not_IE16Conjecture` negates the complete displayed universal inequality. The nine-point witness at degree four uses arbitrary normalized complex polynomials, actual complex norms and *Real.pi*. The full minimum is attained and equals $3003003000/1001003001001$; every five-point subset has an attained minimum below $23/10000$, and their actual maximum is positive. The ratio exceeds $13/10 > 4/\pi$. Exact root coordinates, positive weights and four vanishing moments certify the full optimum. Generic Lagrange interpolation and a finite occupancy argument cover every five-point subset. The stronger unbounded-ratio theorem remains an informal source result outside this formalization; the finite counterexample already settles the original universal target.

The repository ran [successful non-root Ubuntu verification](#) on the pinned canonical proof. All fifteen Comparator statements, permitted axiom closures and Lean's default kernel passed, together with actual sandbox and rejection controls. Every exported theorem reports only *propxet*, *Classical.choice* and *Quot.sound*. [Raw evidence and reproduction commands](#), [statement acceptance](#) and [final canonical referee acceptance](#) preserve the verification scope. AI assistance and independent agent reviews are disallowed; no external human peer review or official Tau Ceti endorsement is claimed.

Original problem (retained)

Let $L \subset \mathbb{C} \setminus \{0\}$ consist of $n \geq 3$ distinct points. For a nonempty $S \subseteq L$, define

$$M_k(S) = \min_{p \in \mathbb{C}[z], \deg p \leq k, p(0)=1} \max_{z \in S} |p(z)|.$$

Is it true, simultaneously for every such L and $1 \leq k \leq n - 2$, that

$$M_k(L) \leq \frac{4}{\pi} \max_{S \subseteq L, |S|=k+1} M_k(S)?$$

The constant is required to be independent of dimension, degree, and the locations of the points.

For a normal matrix with spectrum L , $M_k(L)$ is the largest relative residual norm attainable at GMRES step k , over unit initial residuals. The smaller-set quantities have explicit formulas through Lagrange interpolation. The question therefore asks for a sharp, dimension-independent certificate of worst-case convergence using small subsets of the spectrum. It is distinct from equality of ideal and worst-case GMRES for a nonnormal Jordan block (IE-02).

References

J. Liesen and P. Tichý, *The worst-case GMRES for normal matrices*, BIT 44 (2004), 79–98, conjecture (3.16), pp. 91–92, and Appendix ([author copy](#)). Their *A min-max problem on roots of unity*, TU Berlin Preprint 28-2003, conclusion (9.1), proves selected roots-of-unity cases and sharpness of $4/\pi$ ([university record](#)). Their survey *Convergence analysis of Krylov subspace methods*, GAMM-Mitteilungen 27 (2004), discussion after (11), restates the conjecture ([author copy](#)).

Status check — 2026-09-08

Searched the original title with “conjecture”, “ $4/\pi$ ”, “ $4/\pi$ ”, “proof”, “counterexample”, and 2025–2026; checked the authors’ current publication lists. The later Jordan-block papers concern a different conjecture. No resolution or recent explicit reaffirmation of this particular bound was located; its open-status evidence is historical and bounded.

Audit update — 2026-09-10

The [primary paper](#), §3.2.1, proves the stronger subset equality for every real spectrum, a substantive subclass of the displayed target. Its complex-spectrum conjecture remains in §3.2.2. Searches for the Liesen–Tichý conjecture and later sharp normal-GMRES subset bounds found no general resolution; the current status records the real-spectrum partial result without asserting exhaustive literature coverage.